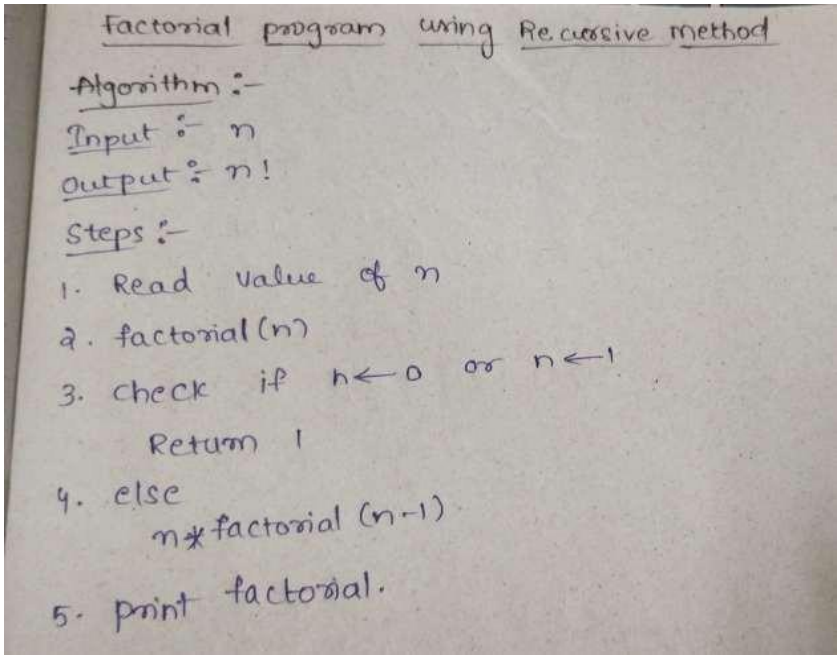
Practical - 04

Aim: Implementation and Time analysis of factorial program using iterative and recursive method

Recursive method -

Algorithms:



Code:-

```
#include <iostream> using namespace std;

int factorial(int n)
{
   if (n == 0 || n == 1)        return 1;

   return n * factorial(n - 1);
}

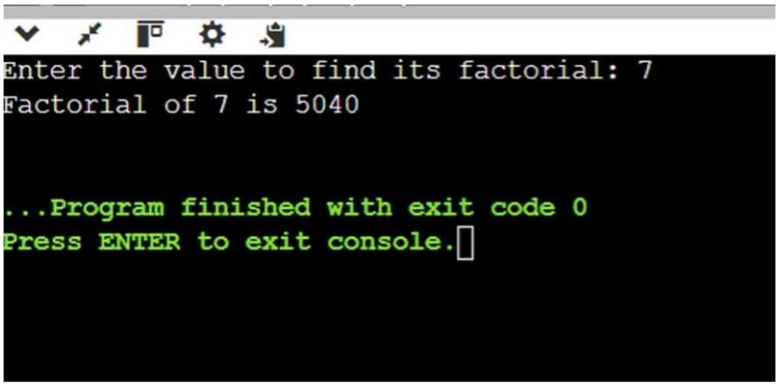
int main()
{    int n;
```

```
    cout << "Enter a number: ";     cin >> n;

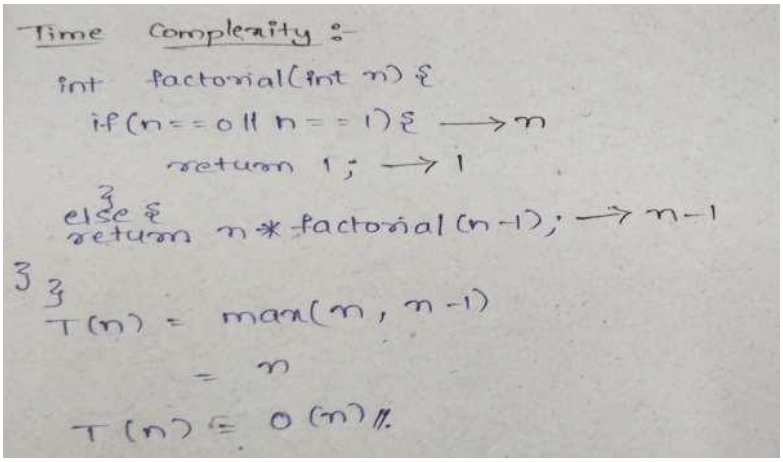
    cout << "Factorial = " << factorial(n) << endl;

    return 0; }
```

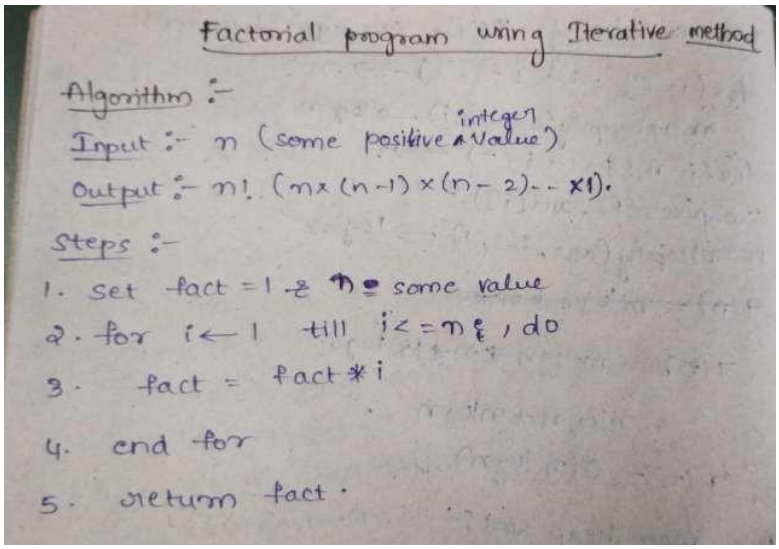
Output:-



Timecomplexity:-



Iterative method - Algorithms:



Code:

```
#include <iostream> using namespace std;

int factorial(int n)
{    int fact = 1;

   for (int i = 1; i <= n; i++)
   {       fact = fact * i;
   }

   return fact;
}

int main() {    int n;    cout << "Enter a number: ";    cin >> n;

   cout << "Factorial = " << factorial(n) << endl;

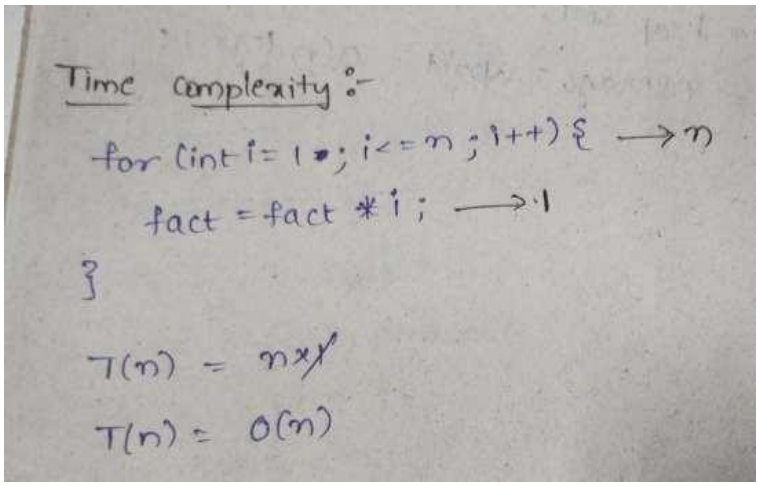
   return 0;
}
```

Output:

```
Enter the value to find its factorial: 7
Factorial of 7 is 5040


...Program finished with exit code 0
Press ENTER to exit console.
```

Time complexity:



Conclusion:-

The factorial program was implemented using both iterative and recursive methods. Both approaches produce the same result, but they differ in their execution and memory requirements.

The iterative method uses a loop to calculate the factorial and has a time complexity of O(n) and a space complexity of O(1). Therefore, it is generally more memory-efficient and suitable for larger values of n.